NEGATIVE EDGE CORRELATION FOR WEIGHTED SPANNING FORESTS OF COMPLETE GRAPHS

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ABSTRACT. Let $\mathbb{P}_t$ be the arboreal gas on the complete graph K_n: the probability measure on acyclic edge sets F proportional to $t^{|F|}$. We prove that for every n and every $t > 0$ any two distinct edges e, f satisfy $\mathbb{P}_t(e, f \in F) < \mathbb{P}_t(e \in F)\,\mathbb{P}_t(f \in F)$. The case $t = 1$ is due to Gupta. For two edges sharing a vertex the inequality follows from the real-rootedness of the degree polynomial of a vertex, which is the characteristic polynomial of a rank-one perturbation of a Hermite Jacobi matrix. For disjoint edges the edge marginal satisfies an explicit Riccati equation, and the covariance is controlled by a barrier argument along it.

1. INTRODUCTION

Let $G = (V, E)$ be a finite graph and $t > 0$. The *arboreal gas* (or weighted spanning forest measure) on G is the probability measure

$$\mathbb{P}_t(F) = \frac{t^{|F|}}{Z_G(t)}, \qquad F \subseteq E \text{ acyclic},$$

where $Z_G(t) = \sum_F t^{|F|}$. It is the $q \to 0$ limit of the random-cluster model with edge weight $p/(1-p) = qt$. Kahn [10], Grimmett and Winkler [7] and Pemantle [11] asked whether the uniform spanning forest ($t = 1$) has negatively correlated edges, as uniform spanning trees do [2, 6]; for the arboreal gas at every $t > 0$ this is Conjecture 1.8 of Bauerschmidt, Crawford, Helmuth and Swan [1]. The conjecture is open for general graphs; for series–parallel graphs the uniform spanning forest satisfies the stronger Rayleigh property [5].

On complete graphs several cases are known. Gupta [8] proved negative correlation of distinct edges of the uniform random forest of K_n for every n, and Cao and Fu [3] did the same for the complete bipartite graphs $K_{m,n}$. Tang and Zhang [16] proved it on K_n for n large for uniform spanning forests with a fixed number of components. Huang [9] proved it for adjacent edges at sufficiently large t on any fixed graph, and on K_n for n sufficiently large when t is sufficiently small or sufficiently large. Çiçeksiz and Ravichandran [4] proved negative edge correlation for the arboreal gas on any finite graph at sufficiently large fugacity, under a nondegeneracy condition, and for the random-cluster model on K_n with $0 < q < 1$ and common edge weight at least one; the arboreal gas is the limit in which the weight tends to zero, so it lies outside that range. Tang and Zhang [12] prove negative edge correlation for the random-cluster model with $0 \le q < 1$ and fixed edge weight on sufficiently large regular graphs of diverging degree satisfying an isoperimetric condition. We prove the complete-graph case at every fugacity.

Theorem 1.1. *Let $n \ge 3$ and $t > 0$. For any two distinct edges e, f of K_n,*

$$\mathbb{P}_t(e \in F,\ f \in F) < \mathbb{P}_t(e \in F)\,\mathbb{P}_t(f \in F).$$

By symmetry there are two cases: e and f share a vertex (Section 3), or they are disjoint, which requires $n \ge 4$ (Sections 4–7). For disjoint edges the proof has three ingredients: exact formulas for the four forest sums in terms of one sequence of Hermite-type polynomials, a Riccati equation for the edge marginal $\rho(t) = \mathbb{P}_t(e \in F)$, and the observation that the covariance is

a fixed quadratic F_t evaluated at $\rho(t)$. Positivity of $F_t(\rho(t))$ is then propagated from both endpoints $t \to 0$ and $t \to \infty$ by a first-zero argument, which needs the sign of the derivative of $F_t(\rho)$ along each root of F_t. These signs come from two polynomial identities and the location of the zeros of a discriminant and of a cubic.

2. Generating functions

Throughout, $a = 1/t$. Let $T(w)$ be the exponential generating function of rooted labelled trees and $U(w)$ that of unrooted labelled trees (both without the empty tree). Then $T = we^T$, $wU' = T$ and $U = T - T^2/2$, since both sides of the last identity vanish at 0 and have derivative T/w (using $(1-T)T' = T/w$).

For $b, a \in \mathbb{R}$ define polynomials $P_k = P_k(b, a)$ by

$$e^{bu - au^2/2} = \sum_{k \geq 0} P_k \, \frac{u^k}{k!}.$$

Differentiating in u gives $P_0 = 1$, $P_1 = b$ and

$$P_k = bP_{k-1} - a(k-1)P_{k-2} \qquad (k \geq 2). \tag{1}$$

Put $Q_N = P_N - NP_{N-1}$.

Lemma 2.1. *For every power series g and every $N \geq 0$,*

$$[w^N]\, e^{aU(w) + zT(w)} g(T(w)) = [u^N]\, (1-u)\, g(u)\, e^{(N+a+z)u - au^2/2}.$$

Proof. Write the coefficient as a residue in w and substitute $w = ue^{-u}$, so that $T = u$, $U = u - u^2/2$ and $dw/w^{N+1} = (1-u)e^{Nu}\, du/u^{N+1}$. This is the Lagrange inversion formula as a formal identity. □

The degree polynomial of a vertex. Fix a vertex v of K_n, put $N = n-1$, and let $H(z) = \sum_F t^{|F|} z^{\deg_F(v)}$. Removing v from a forest leaves a forest on the other N vertices in which every component is joined to v by at most one edge. A component on k vertices that is not joined to v has weight t^{k-1}; a component joined to v through a chosen vertex has weight $t^{k-1} \cdot tz = zt^k$. By the exponential formula and Lemma 2.1 with $g = 1$,

$$H(z) = N!\, t^N [w^N]\, e^{aU(w) + zT(w)} = t^N\, Q_N(N + a + z,\, a). \tag{2}$$

In particular $Z := Z_{K_n}(t) = H(1) = t^{n-1} Q_{n-1}(n + a, a)$.

3. Edges with a common vertex

Lemma 3.1. *All zeros of H are real and negative, and they are not all equal when $N \geq 2$.*

Proof. Let J_N be the $N \times N$ symmetric tridiagonal matrix with zero diagonal and off-diagonal entries $\sqrt{a}, \sqrt{2a}, \ldots, \sqrt{(N-1)a}$, and let e_N be the last coordinate vector. Expanding along the last row and using (1) gives $\det(bI - J_N) = P_N(b, a)$. The determinant of $bI - J_N - Ne_N e_N^\top$ is affine in its last diagonal entry, with cofactor $\det(bI - J_{N-1}) = P_{N-1}$, so

$$\det(bI - J_N - Ne_N e_N^\top) = P_N(b, a) - NP_{N-1}(b, a) = Q_N(b, a).$$

Hence by (2) the zeros of H are $\lambda - N - a$ for the eigenvalues λ of the real symmetric matrix $J_N + Ne_N e_N^\top$, and are real. The coefficients of H are nonnegative and $H(0) \geq 1$, so H has no zero in $[0, \infty)$. For $N \geq 2$ this matrix has a nonzero off-diagonal entry, so it is not scalar and its eigenvalues are not all equal. □

Proposition 3.2. *For $n \geq 3$, $t > 0$ and distinct edges e, f with a common vertex, $\mathbb{P}_t(e, f \in F) < \mathbb{P}_t(e \in F)\mathbb{P}_t(f \in F)$.*

Proof. The leading coefficient of H is t^N (the star at v), so by Lemma 3.1 $H(z) = t^N \prod_{j=1}^{N}(z+\lambda_j)$ with $\lambda_j > 0$ not all equal. Let $Y = \deg_F(v)$ and $p_j = 1/(1+\lambda_j)$. Logarithmic differentiation at $z = 1$ gives

$$\mu := \mathbb{E}_t Y = \sum_j p_j, \qquad \mathbb{E}_t[Y(Y-1)] = \frac{H''(1)}{H(1)} = \mu^2 - \sum_j p_j^2.$$

Since the p_j are not all equal, $\sum_j p_j^2 > \mu^2/N$, and so $\mathbb{E}_t[Y(Y-1)] < (N-1)\mu^2/N$. The N edges at v are exchangeable; with $\rho = \mathbb{P}_t(e \in F)$ and $\eta = \mathbb{P}_t(e, f \in F)$ we have $\mu = N\rho$ and $\mathbb{E}_t[Y(Y-1)] = N(N-1)\eta$. Hence $\eta < \rho^2$. □

4. Forest sums with prescribed edges

From now on $n \geq 4$, $a = 1/t$, $b = n + a$, and all P_k are evaluated at (b, a). Let E, L and D be the sums of $t^{|F|}$ over forests containing a fixed edge, two fixed edges with a common vertex, and two fixed disjoint edges, respectively, and put $\rho = E/Z$ and $h = (n-2)(n-3)$.

Lemma 4.1.

$$Z = t^{n-1}Q_{n-1}, \quad E = t^{n-1}Q_{n-2}, \quad L = t^{n-1}Q_{n-3}, \quad D = t^{n-1}\big(P_{n-3} - (n-4)P_{n-4}\big).$$

Proof. The formula for Z is (2) at $z = 1$. Contract a prescribed edge uv to a vertex m. A forest containing uv corresponds to a forest of the contracted graph in which each edge from m to an ordinary vertex x stands for one of the two edges ux, vx; at most one of them can be present. So m is a vertex of degree fugacity $z = 2$ among $n-2$ ordinary vertices, and (2) with $N = n-2$, $z = 2$, times the weight t of uv, gives E. Contracting two prescribed edges with a common vertex gives a vertex of fugacity 3 among $n-3$ ordinary vertices, and L in the same way.

For two disjoint prescribed edges, contract each to a vertex, m_1 and m_2, each of fugacity 2, among $N = n-4$ ordinary vertices; the edge m_1m_2 stands for four edges and has weight $4t$. If m_1, m_2 lie in different components, the components are counted by e^{aU+4T}. If they lie in one component, the path between them has $k \geq 0$ ordinary internal vertices; its edges have total weight $4t^{k+1}$ and each internal vertex with the trees hanging from it contributes T/t, giving $4tT^k$. Hence the contracted sum is $N!\,t^N[w^N]e^{aU+4T}\big(1 + 4t/(1-T)\big)$, and by Lemma 2.1,

$$D = t^{n-2}\big((1+4t)P_{n-4} - (n-4)P_{n-5}\big) = t^{n-1}\big(P_{n-3} - (n-4)P_{n-4}\big),$$

the second form by (1) with $k = n-3$. For $n = 4$ the term with P_{-1} is absent and both forms equal $t^2(1+4t)$. □

Write $P = P_{n-1}$ and $R = P_{n-2}$. Eliminating P_{n-3}, P_{n-4} with (1) expresses the four quantities of Lemma 4.1, divided by t^{n-1}, as

$$P - (n-1)R, \quad \frac{P - nR}{a}, \quad \frac{nP - (n^2+2a)R}{a^2(n-2)}, \quad \frac{(n-4)(n-2)aR + (a - n(n-4))(bR - P)}{a^2 h}.$$

Collecting the coefficients of P and R then verifies the identities

$$\frac{L}{Z} = \frac{(2+nt)\rho - 2t}{n-2}, \tag{3}$$

$$\frac{D}{Z} = \frac{1 - 2(n-4)t + \big(-1/t + 2n - 9 + n(n-4)t\big)\rho}{h}. \tag{4}$$

Lemma 4.2 (Riccati equation)**.** *For $n \geq 4$ and $t > 0$,*

$$2t^2\rho' = t - 2nt^2 + B\rho - n(n-1)t\rho^2, \qquad B = n^2t^2 + (2n+1)t - 1.$$

Proof. Let $M = \binom{n}{2}$. Since $t\,\partial_t$ applied to $\sum_F t^{|F|}$ counts edges, exchangeability gives $tZ' = ME$. Applied to E it counts, besides the prescribed edge, the $2(n-2)$ edges sharing a vertex with it and the $h/2$ disjoint ones, so $tE' - E = 2(n-2)L + hD/2$. Hence $t\rho' = \rho + 2(n-2)L/Z + hD/(2Z) - M\rho^2$, and (3)–(4) give the claim. □

Define

$$\delta = th, \quad \alpha = t - 2(n-4)t^2, \quad \beta_0 = -1 + (2n-9)t + n(n-4)t^2, \quad F_t(r) = \delta r^2 - \beta_0 r - \alpha.$$

By (4), $F_t(\rho) = \delta(\rho^2 - D/Z)$. So, writing $f(t) = F_t(\rho(t))$, the disjoint case of Theorem 1.1 is the statement that $f > 0$ on $(0, \infty)$.

5. The two ends

Lemma 5.1. *For each $n \geq 4$, $f(t)/t^5 \to 2h$ as $t \to 0^+$ and $f(t) \to 4h/n^4$ as $t \to \infty$.*

Proof. Write $Z = \sum Z_k t^k$ and similarly for E, D. Counting edge sets with k edges and removing those containing a cycle,

$$Z_0 = 1,\ Z_1 = M,\ Z_2 = \tbinom{M}{2}; \quad E_1 = 1,\ E_2 = M-1,\ E_3 = \tbinom{M-1}{2} - (n-2);$$

$$D_2 = 1,\ D_3 = M-2,\ D_4 = \tbinom{M-2}{2} - 2(n-2) - 2,$$

with $E_0 = D_0 = D_1 = 0$. For E_3 the excluded sets are the $n-2$ triangles through the prescribed edge; for D_4 they are the $2(n-2)$ triangles through one prescribed edge completed by the other, and the two 4-cycles through both. Multiplying out, $E^2 - ZD = 2t^4 + O(t^5)$, and $\rho = t + O(t^2)$, so $f = th(E^2 - ZD)/Z^2 = 2ht^5 + O(t^6)$.

As $t \to \infty$, ρ is a rational function of t with $\rho \to 2/n$, the edge density of a uniform spanning tree. Write $\rho = 2/n + r_1/t + r_2/t^2 + O(t^{-3})$. The coefficients of t and of t^0 in Lemma 4.2 give $1 + 6/n + n^2 r_1 = 0$ and $-2r_1 = n^2 r_2 + (5-2n)r_1 - 2/n$, so $r_1 = -(n+6)/n^3$ and $r_2 = (42-5n)/n^5$. Inserting these into (4) gives $\rho^2 - D/Z = 4/(n^4 t) + O(t^{-2})$, hence $f \to 4h/n^4$. □

6. Polynomial facts

Let $n \geq 7$ and

$$\varphi(t,r) = t - 2nt^2 + Br - n(n-1)tr^2, \qquad G(t,r) = 2t^2\partial_t F_t(r) + \partial_r F_t(r)\,\varphi(t,r).$$

By Lemma 4.2, $f'(t) = G(t, \rho(t))/(2t^2)$. Put $A = n(n-4)$,

$$\mathcal{L} = n^3(n-4)t^4 + 2n(n^2-4n-6)t^3 + 3(n^2-9n+3)t^2 + 2(n^2-9n+9)t - (2n-3),$$

$$\mathcal{C} = 2n^2(n-4)t^3 + 3n(n-6)t^2 + (2n^2-16n+15)t - (2n-3),$$

$$J = 2A^2t^3 - 6A(n-2)t^2 + 6(n-1)(n-3)t - (2n-3).$$

Lemma 6.1. *As polynomials in n, t, r,*

$$hG = F_t(r)\big[-2hn(n-1)tr + 2ht + 2hB - n(n-1)\beta_0\big] + 2(\mathcal{L}r - t\mathcal{C}), \tag{5}$$

$$\delta t^2\mathcal{C}^2 - \beta_0 t\mathcal{L}\mathcal{C} - \alpha\mathcal{L}^2 = -h^2t^5 J. \tag{6}$$

Proof. Both are verified by expanding the definitions; the accompanying scripts do so in exact arithmetic. □

Let $\ell = 2\mathcal{L}/h$ and $c_0 = -2t\mathcal{C}/h$. By (5), $G(t,r) = \ell r + c_0$ at every real root r of F_t. Let $\Delta = \beta_0^2 + 4\delta\alpha$ be the discriminant of F_t. When $\Delta \geq 0$ write $r_- \leq r_+$ for the roots and $g_\pm(t) = G(t, r_\pm(t))$. Since $r_- + r_+ = \beta_0/\delta$ and $r_- r_+ = -\alpha/\delta$, identity (6) gives

$$g_- g_+ = -\frac{4t^4 J}{h}. \tag{7}$$

The leading coefficient of ℓ in t is $2n^3(n-4)/h > 0$.

Lemma 6.2. *For $n \geq 7$, J has exactly one positive zero β. It satisfies $\beta > 1/(n-3)$, $J < 0$ on $(0,\beta)$, $J > 0$ on (β,∞) and $J'(\beta) > 0$.*

Proof. One checks $J'(t) = 6\big[(At-(n-2))^2 - 1\big]$, with positive zeros $t_a = (n-3)/A$ and $t_b = (n-1)/A$, and $J(t_a) = -(n-6)/(n-4) < 0$, $J(t_b) = -(n-2)/n < 0$, $J(0) = -(2n-3) < 0$. So J is negative on $[0,t_b]$ and strictly increasing on (t_b,∞) to $+\infty$. Finally $t_b > 1/(n-3)$ because $(n-1)(n-3) - n(n-4) = 3$. □

Lemma 6.3. *For $n \geq 7$, let $t_c = 1/(n-3)$. Then Δ has exactly two positive zeros $\tau_1 < t_c < \tau_2$, both simple; $\Delta > 0$ on $(0,\tau_1) \cup (\tau_2,\infty)$, $\Delta < 0$ on (τ_1,τ_2), $\Delta'(\tau_1) < 0$ and $\Delta'(\tau_2) > 0$.*

Proof. Expanding,

$$\Delta = 1 + (18-4n)t + (6n^2-48n+105)t^2 + (-4n^3+38n^2-136n+192)t^3 + n^2(n-4)^2t^4.$$

Let $c = n-3 \geq 4$. Under $t = u/(c(1+u))$, which maps $(0,\infty)$ onto $(0,t_c)$, the polynomial $c^4(1+u)^4\Delta$ has coefficients, in increasing powers of u,

$$c^4,\ 6c^3,\ c^2(6c+15),\ -2c(c+1)(2c-9),\ -3(c-3)(c+1).$$

The sequence has exactly one sign change, so by Descartes' rule there is exactly one positive zero, and it is simple. Under $t = (1+u)/c$, which maps $(0,\infty)$ onto (t_c,∞), the coefficients of $c^4\Delta$ are

$$-3(c-3)(c+1),\ 2(2c^3-13c^2+3c+18),\ 9(2c^3-5c^2-2c+6),\ 6(c-1)(3c^2-c-6),\ (c+3)^2(c-1)^2,$$

and with $c = x+4$ the third and fourth are positive, since $2c^3-5c^2-2c+6 = 2x^3+19x^2+54x+46$ and $3c^2-c-6 = 3x^2+23x+38$. Again there is exactly one sign change, and one simple positive zero. At $t = t_c$, $\Delta = -3(c-3)(c+1)/c^4 < 0$. The signs at 0, at t_c and at ∞ give the rest. □

7. The barrier argument

Proposition 7.1. *For $n \geq 4$, $t > 0$ and disjoint edges e, f, $\mathbb{P}_t(e,f \in F) < \mathbb{P}_t(e \in F)\mathbb{P}_t(f \in F)$.*

Proof. For $n = 4, 5, 6$, Lemma 4.1 gives

$$E^2 - ZD = t^4(2+4t),$$
$$E^2 - ZD = t^4(2+24t+84t^2+100t^3),$$
$$E^2 - ZD = t^4(2+48t+462t^2+2172t^3+5184t^4+5184t^5),$$

which are positive for $t > 0$. Let $n \geq 7$; we show $f > 0$ on $(0,\infty)$.

Signs at the double roots. Write $F_t(r) = \delta\big(r - \beta_0/(2\delta)\big)^2 - \Delta/(4\delta)$. At a zero τ of Δ the double root is $r^* = \beta_0/(2\delta)$, where $\partial_r F = 0$ and

$$G(\tau, r^*) = 2\tau^2 \partial_t F_\tau(r^*) = -\frac{\tau^2 \Delta'(\tau)}{2\delta(\tau)}.$$

By Lemma 6.3 this is positive at τ_1 and negative at τ_2. At τ_2, (7) gives $0 < G(\tau_2, r^*)^2 = -4\tau_2^4 J(\tau_2)/h$, so $J(\tau_2) < 0$ and $\tau_2 < \beta$ by Lemma 6.2.

The interval $(0,\tau_1]$. Here $t < t_c < \beta$, so $g_- g_+ > 0$ by (7); neither factor vanishes, and both tend to the positive value $G(\tau_1, r^*)$ at τ_1. Hence $g_\pm > 0$ on $(0,\tau_1]$. By Lemma 5.1, $f > 0$ near 0. If f had a first zero $t_0 \leq \tau_1$, then $f'(t_0) \leq 0$; but $\rho(t_0)$ is a root of F_{t_0}, so $f'(t_0) = g_\pm(t_0)/(2t_0^2) > 0$. Thus $f > 0$ on $(0,\tau_1]$.

The interval (τ_1,τ_2). Here $\Delta < 0$ and $\delta > 0$, so $F_t > 0$ on all of $\mathbb{R}$, and $f > 0$.

The lower root above τ_2. We show $g_- < 0$ on $[\tau_2, \infty)$. On (τ_2, β), $g_- g_+ > 0$, and both factors are negative at τ_2, so $g_\pm < 0$ on $[\tau_2, \beta)$. For $t > \beta$, $g_- g_+ < 0$, and by (5)

$$g_+ - g_- = \ell\,(r_+ - r_-). \tag{8}$$

If $\ell(t) = 0$ for some $t > \beta$ then $g_+ = g_-$ and $g_- g_+ \geq 0$, which is false; as ℓ is eventually positive, $\ell > 0$ on (β, ∞). At β, $\Delta > 0$ since $\beta > \tau_2$, so $g_\pm$ are differentiable near β, and $g_- g_+$ has a simple zero there with derivative $-4\beta^4 J'(\beta)/h < 0$. If $\ell(\beta) = 0$ then $g_-(\beta) = g_+(\beta) = 0$ and the derivative of $g_- g_+$ at β would vanish; so $\ell(\beta) > 0$. By continuity $g_\pm(\beta) \leq 0$, their product is 0 and their difference is positive, so $g_-(\beta) < 0 = g_+(\beta)$. For $t > \beta$, (8) and $g_- g_+ < 0$ give $g_- < 0 < g_+$.

The interval $[\tau_2, \infty)$. Since $\rho \leq 2/n$ (a forest has at most $n-1$ edges) while $r_+(t) \to \infty$, and $f > 0$ for large t by Lemma 5.1, we have $\rho < r_-$ for all large t. Suppose f vanishes somewhere in $[\tau_2, \infty)$, and let t_0 be its largest zero there. On (t_0, ∞) the roots are real and distinct and $f \neq 0$, so $\rho < r_-$ throughout, and $\rho(t_0) = r_-(t_0)$. Positivity of f to the right of t_0 requires $f'(t_0) \geq 0$, while $f'(t_0) = g_-(t_0)/(2t_0^2) < 0$. Hence $f > 0$ on $[\tau_2, \infty)$. □

Theorem 1.1 follows from Propositions 3.2 and 7.1; for $n = 3$ every pair of edges has a common vertex.

Remark 7.2. The argument uses only the exact counts of Lemma 4.1 and properties of explicit polynomials. Whether the same method applies to other vertex-transitive graphs, or to unequal edge weights, is not addressed here.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

OpenAI Codex systems developed the mathematical argument and the exact verification code. Anthropic Claude AI systems audited the proofs, rechecked the algebra with independent code, compared the result with the literature, and wrote and prepared this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for publication rests with the author. The AI-assisted internal reviews and compilation checks are not external peer review or formal verification.

Data and code availability

The scripts that verify Lemmas 4.1–6.3 and the base cases, and that check Theorem 1.1 by exhaustive enumeration for $n \leq 7$, are in the repository `https://github.com/yspennstate/kn-forest-negative-correlation`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [13].

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